

KAVIPRAKASH RAMALINGAM

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EDUCATION

Northeastern University

Master of Science in Information Systems

Sep. 2023 – May 2025

Boston, MA

EXPERIENCE

Renault Nissan

April 2021 – August 2023

Senior Fullstack Engineer

- Led an Agile team of 6 for Renault's ISIT systems, managing **1,500** global supplier sites across **25** countries.
- Redesigned invoice processing systems for Renault's SDV entity, AMPERE, with **Spring Boot** RESTful APIs for data processing, integrating automated **ETL** pipelines that reduced manual interventions by **50%** and achieved **99%** data validation accuracy.
- Helped architect a government-access system for Renault Romania by implementing an **OAuth2**-compliant authentication and authorization server, scaling from **100,000** to over **1 million** tokens issued daily.
- Engineered performant web applications supporting Renault's EV strategy using **React**, **Tailwind CSS**, and Dropwizard-based **Java** microservices, optimizing **Redux** state management and reducing frontend latency by **40%**.

Capgemini Technologies

January 2018 – March 2021

Associate Consultant

- Automated infrastructure provisioning with Terraform and Ansible, cutting setup time by **40%**.
- Designed a CI/CD pipeline for deploying microservices on **AWS** using **Jenkins**, **Git**, and automated EC2 patching via AWS Systems Patch Manager, ensuring up-to-date instances.
- Introduced **Prometheus** and **Grafana** for advanced monitoring, decreasing issue resolution time by **35%** and improving system reliability.

PROJECTS

WebVectorAI | Python, OpenAI, Langchain

December 2024

- Built an application leveraging web scraping to extract and vectorize website content into a 1536-dimensional format for storage in a vector database.
- Integrated Langchain and OpenAI GPT-3.5 Turbo LLM for conversational AI capabilities.

Rehabilitation Games & Extended Reality (ReGame-XR) | Python, YOLOv8, Unity

June 2024

- Developed an AI pipeline in Python to visualize gaze points from raw eye-tracking data, aiding autism behavior analysis
- Trained a YOLOv8 object detection model to analyze gaze patterns, distinguishing differences between autistic and neurotypical individuals.
- Assisted in developing a Unity-based GTA game to record and visualize gaze points, enabling users to explore different perspectives for autism behavior analysis.

PULSAR - Digital Transformation | React, Node, MongoDB, SwiftUI, Firebase, PWA

September 2023

- A vehicle monitoring web app using React, Node.js, Express, and MongoDB with a microservices architecture. A vehicle health monitoring iOS app, leveraging SwiftUI, Firebase, and MapKit for seamless geolocation and dynamic UI.
- Strengthened security using OAuth 2.0, JWT, and Firebase Authentication.
- Improved performance: - Web app: Optimized Redux Toolkit, server-side rendering (SSR), lazy loading, and database queries. iOS app: Integrated CoreLocation, real-time Firebase Firestore queries, and image uploads via Imgur API.
- Designed user-centric features: - Web: Added PWA capabilities and CI/CD pipelines with GitHub Actions deployed on AWS EC2. iOS: Developed an inventory-based parts search and a feedback system for user contributions.

HORSE & AMPERE – Powertrain | React, Spring Boot, Jenkins, Docker, Jest

November 2022

- Engineered a scalable web application to oversee powertrain production, utilizing React for UI and Spring Boot for backend services, improving operational efficiency by 25%.
- Developed & integrated RESTful APIs, facilitating seamless data exchange between systems, reducing data retrieval by 40%.
- Applied DevOps practices with CI/CD pipelines (Jenkins) and Docker, reducing deployment times by 30% while supporting a 50% user base increase without performance degradation.
- Led the development of a real-time analytics dashboard for electric vehicle monitoring using React and Node.js, enhancing data processing speed by 20%.
- Conducted comprehensive testing with Jest and Mocha, achieving 95% test coverage and reducing post-deployment defects.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript (ES6+), C++, Python, TypeScript, Swift

Certifications: GCP Certified Cloud Practitioner, DevOps Certified

Databases (SQL & NoSQL): SQLite, MySQL, MongoDB, PostgreSQL, Memcached, Firebase

Frontend Technologies: Bootstrap, CSS3, HTML5, ReactJS, Next.js, Socket.IO, Redux, SASS, Tailwind CSS, UIKit, SwiftUI

Software/Tools: AWS, Docker, Kubernetes, Git, GitHub Actions, Jenkins, Terraform, ELK, Grafana, Apache ZooKeeper, JIRA, Confluence

Libraries/Frameworks: TensorFlow, PyTorch, NLP, Machine Learning Algorithms, Pandas, NumPy, Scikit-learn, SciPy, LangChain, Streamlit, Hugging Face Transformers, Selenium

Java Ecosystem: Kafka, Hibernate, JPA, OpenAPI, Spring Boot, Spring Security, Tomcat

iOS Development: Swift, Xcode IDE, Core Data, SwiftUI, iOS Simulator